CHAPTER 12

HUMAN CAPITAL FORMATION IN INDIA

CHAPTER ESSENCE AT A GLANCE:
Examine human development: physical versus human capital, five sources of human capital formation, economic growth linkages, Right to Education Act, and institutional roles of NCERT, UGC, and AICTE.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Physical vs Human Capital Duality:** Physical capital (machinery, buildings) is tangible and depreciates with time; Human capital (skills, knowledge, health) is embodied in human beings and increases productivity.

• **Five Sources of Human Capital Formation:** Expenditure on Education, Expenditure on Health (preventive, curative, social medicine), On-the-job Training, Expenditure on Migration, and Expenditure on Information.

• **Human Capital vs Human Development:** Human capital views human beings as instruments to enhance labor productivity; Human development views human well-being, education, and health as valuable ends in themselves.

• **Statutory Educational Milestones:** The 86th Constitutional Amendment 2002 and Right to Education (RTE) Act 2009 made free and compulsory elementary education a fundamental right for children aged 6 to 14.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Human Capital Formation (HCF)} = \text{Education} + \text{Health} + \text{On-the-Job Training} + \text{Migration} + \text{Information}$$

Five Primary Sources of Human Capital Formation

$$\text{Economic Growth Synergy} \Longleftrightarrow \text{Human Capital Investment} \rightarrow \uparrow \text{Labor Productivity} \rightarrow \uparrow \text{Real GDP Growth}$$

Endogenous Growth Productivity Multiplier

$$\text{RTE Mandate} \Longleftrightarrow \text{Free ¤ext\& Compulsory Education for Children Aged 6 to 14 Years}$$

86th Constitutional Amendment Statutory Right

$$\text{Health Investment} = \text{Preventive (Vaccines)} + \text{Curative (Medicines)} + \text{Social (Clean Water)}$$

Tripartite Healthcare Expenditure Dimensions

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Human Capital treats education and health as a means to increase economic output; Human Development treats education and health as essential human rights and ends in themselves.

• **Alert:** Expenditure on migration is an investment in human capital because the increased income earned in urban areas exceeds the costs of migration.

• **Alert:** Tapas Majumdar Committee (1998) estimated that India needed an investment of around Rs 1.37 lakh crore over 10 years to universalize school education.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Physical Capital and Human Capital with one example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Physical Capital:** Tangible, manufactured productive assets that can be easily bought, sold, and separated from owners (e.g. factory machines, trucks). **[1 Mark]**

• **Human Capital:** Intangible skills, health, and intellectual competencies embodied in human beings that cannot be separated from the individual (e.g. software engineer skills, doctor expertise). **[1 Mark]**

2. Distinguish between Human Capital and Human Development. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Human Capital:** Considers education and health as a means to increase worker productivity and maximize economic output. **[1 Mark]**

• **Human Development:** Based on the philosophy that education and health are integral to human well-being and valuable ends in themselves, regardless of productivity. **[1 Mark]**

3. Explain any three major sources of Human Capital Formation. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Investment in Education:** Enhances intellectual capabilities, raises future earning potential, and accelerates modernization. **[1 Mark]**

• **Investment in Health:** Directly increases physical stamina, reduces sick absenteeism, and raises daily labor output. **[1 Mark]**

• **On-the-job Training:** Fast-tracks technical skill acquisition on factory floors, raising labor productivity beyond the costs of training. **[1 Mark]**

4. Why are Expenditure on Migration and Expenditure on Information considered sources of Human Capital Formation? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Migration Expenditure:** People migrate from rural villages to cities in search of higher salaries; the extra earnings obtained far exceed the transit and living costs of migration. **[1.5 Marks]**

• **Information Expenditure:** Spending money to acquire information about labor market wages, jobs, and educational courses prevents misallocation of human talent. **[1.5 Marks]**

5. Explain the role of Government regulatory bodies in education and health: NCERT, UGC, AICTE, and ICMR. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **NCERT:** Prepares school curricula, textbooks, and sets standards for primary and secondary school education in India. **[0.75 Mark]**

• **UGC:** Regulates university education, disburses development grants, and maintains standards in higher degree education. **[0.75 Mark]**

• **AICTE:** Regulates and accredits technical, engineering, and management professional institutions. **[0.75 Mark]**

• **ICMR:** Guides and funds biomedical and health science research across medical institutions. **[0.75 Mark]**

6. State the key provisions of the Right to Education (RTE) Act 2009. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Fundamental Right:** Enacted under Article 21A, guaranteeing free and compulsory education to all children aged 6 to 14 years. **[1.5 Marks]**

• **Private School Quota:** Mandates 25 percent reservation of entry-level seats in private non-minority schools for children from economically weaker and disadvantaged sections. **[1.5 Marks]**

7. Discuss the major challenges confronting Human Capital Formation in India: Rapid population growth, Brain drain, Gender disparity, and Inadequate public health infrastructure. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Rapid Population Growth:** Strains existing schools, hospitals, and sanitation facilities, diluting per capita social infrastructure availability. **[1 Mark]**

• **Brain Drain:** Highly trained doctors, scientists, and engineers educated at subsidized public cost emigrate to developed nations for higher wages. **[1 Mark]**

• **Gender Inequality:** Female literacy (around 70 percent) and higher education enrollments lag significantly behind male counterparts. **[1 Mark]**

• **Inadequate Public Health Funding:** Public health spending has remained around 1 to 2 percent of GDP, forcing poor households to bear crippling out-of-pocket medical debts. **[1 Mark]**